

Prometheus AI

Financial Attack Simulation & Early Warning System for AiTM + BEC Attacks

Proposal for AURORA 2026

Team Details

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1. Introduction

Modern financial cyberattacks increasingly use advanced phishing methods such as Adversary-in-the-Middle (AiTM) attacks and Business Email Compromise (BEC) to bypass authentication systems and manipulate financial workflows.

Prometheus AI is an agentic cybersecurity platform designed to simulate, analyze, and predict financial phishing attack progression before financial damage occurs. Instead of only detecting malicious emails or links, the system reconstructs the full attack chain from phishing entry to payment manipulation using AI agents, sandbox analysis, and attack propagation modeling.

The platform aims to provide organizations with an early warning system capable of detecting sophisticated financial threats before unauthorized transactions are completed.

2. Problem Statement

Modern phishing attacks have evolved beyond simple credential theft. Attackers now use:

- Reverse-proxy phishing websites
- Real-time MFA interception
- Session hijacking
- Internal email manipulation
- Payment redirection techniques

These attacks are highly dangerous because attackers operate inside legitimate authenticated sessions, making detection difficult for traditional security systems.

Organizations affected include:

- Government treasury departments
- Banking institutions
- Enterprise finance teams
- Supplier payment systems

Common examples include:

- AiTM attacks that relay credentials and MFA tokens in real time
- BEC attacks that manipulate invoice or payment instructions

Most existing cybersecurity tools focus on:

- URL filtering
- Malware detection
- Spam prevention
- Reputation analysis

However, they do not simulate how attackers move from phishing entry to financial compromise. This creates a major security gap in detecting multi-stage financial attacks.

An AI-based solution is necessary because modern attacks involve:

- Dynamic attacker behavior
- Human manipulation strategies
- Multi-stage attack progression
- Real-time interaction analysis

Prometheus AI addresses this problem by combining AI reasoning, sandbox execution, and attack propagation simulation to predict financial attack outcomes before financial transactions are compromised.

3. AI Architecture

3.1 Overall System Design

Prometheus AI consists of:

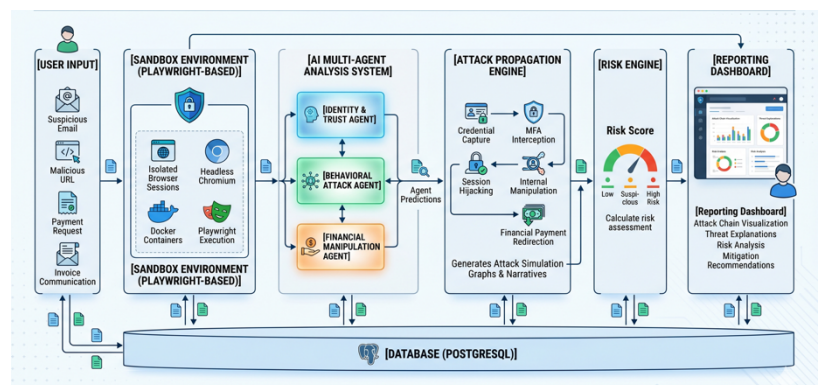
- Secure sandbox browser environment
- AI multi-agent analysis system
- Attack propagation engine
- Financial risk scoring engine
- Explainable reporting dashboard

The system accepts suspicious:

- Emails
- URLs
- Payment requests

Outputs include:

- Risk scores
- Attack simulation graphs
- Early warning alerts
- Mitigation recommendations



3.2 System Workflow

Step 1 : Suspicious Content Submission

Users submit suspicious emails, links, or payment requests.

Step 2 : Sandbox Execution

The interaction executes inside a secure Playwright-based sandbox environment.

Step 3 : AI Agent Analysis

AI agents analyze:

- Identity impersonation
- Behavioral attack patterns
- Financial manipulation intent

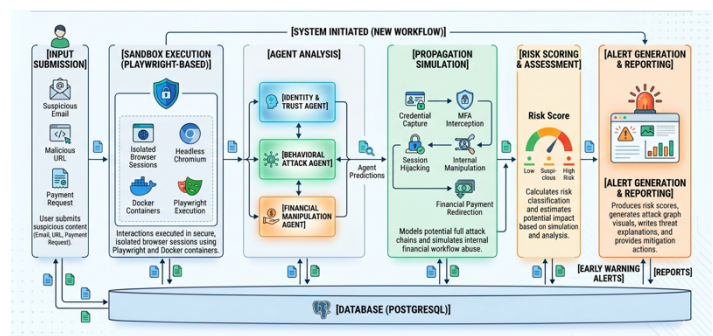
Step 4 : Attack Propagation Simulation

The system reconstructs the possible attack lifecycle from phishing entry to financial compromise.

Step 5 : Risk Assessment & Reporting

The platform generates:

- Risk classifications
- Attack narratives
- Early warning alerts
- Suggested mitigations



3.3 AI Multi-Agent System

Identity & Trust Agent

Detects:

- Domain impersonation
- Reverse-proxy phishing
- Suspicious SSL certificates
- Fake identity indicators

Outputs:

- Identity risk score
- Impersonation indicators

Behavioral Attack Agent

Analyzes:

- Credential relay behavior
- Session hijacking indicators
- Redirect chains
- Suspicious scripts

Outputs:

- AiTM likelihood score

- Behavioral anomaly logs

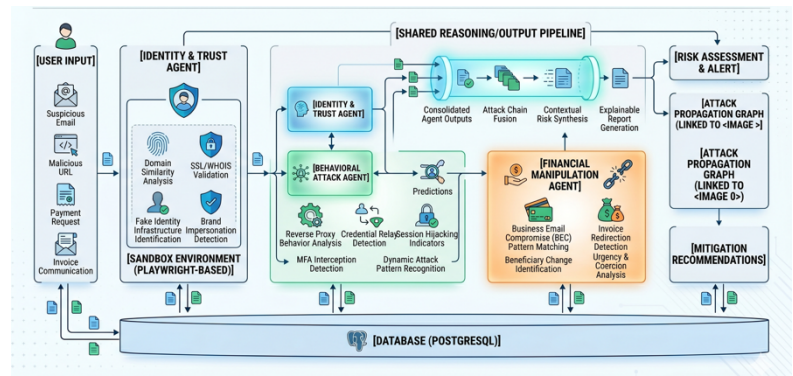
Financial Manipulation Agent

Detects:

- Payment modification attempts
- Beneficiary changes
- Urgency-based manipulation
- Financial coercion patterns

Outputs:

- BEC risk classification
- Financial manipulation indicators



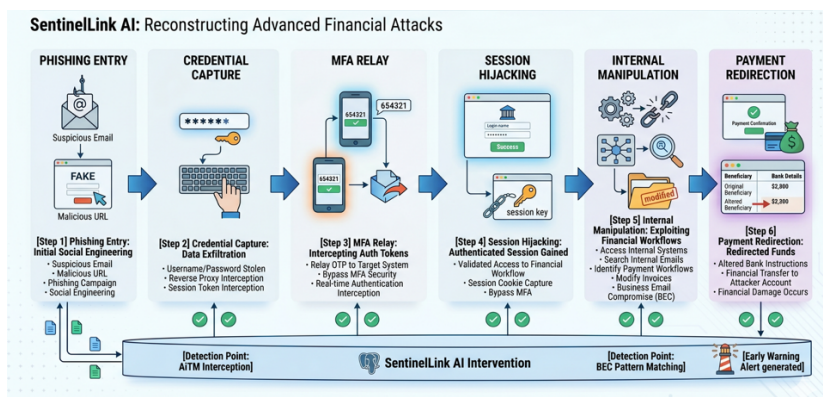
3.4 Attack Propagation Engine

The Attack Propagation Engine is the core innovation of Prometheus AI.

Instead of generating simple phishing alerts, the system simulates:

1. Credential capture
2. MFA relay attacks
3. Session hijacking
4. Internal communication manipulation
5. Financial transaction redirection

The platform produces a structured attack progression model that predicts how attackers may exploit financial workflows.



3.5 Technologies & Frameworks

Frontend: Next.js, Tailwind CSS

Backend: FastAPI

Sandbox Environment: Playwright, Docker, Headless Chromium

AI Layer: OpenAI APIs, Google APIs, Agent-based orchestration

Database: PostgreSQL

4. Ethical Safeguards

Prometheus AI includes multiple ethical and security protections:

- Sandbox-isolated testing environment
- Encrypted communication and secure API handling
- Restricted data retention
- Human supervision for high-risk decisions
- Explainable AI-generated attack narratives
- Controlled simulations without real-world exploitation

The system is designed strictly for defensive cybersecurity analysis and does not automate offensive attacks.

Limitations include:

- Possible false positives
- Simplified sandbox simulations in MVP stage
- No replacement for enterprise security infrastructure

5. Expected Impact & Innovation

Expected Impact

Prometheus AI can help organizations:

- Detect sophisticated phishing attacks earlier
- Reduce financial fraud risks
- Improve cybersecurity awareness
- Strengthen financial workflow protection
- Improve incident investigation efficiency

The platform can also be used for:

- Cybersecurity education
- Attack simulation training
- Security awareness programs

Future improvements may include:

- Enterprise-scale deployment
- Real banking integrations
- Automated mitigation workflows
- Continuous AI learning systems

Innovation

Key innovative features include:

- AI-driven attack propagation modeling
- Combined AiTM and BEC analysis
- Financial workflow simulation
- Explainable cybersecurity reasoning
- Pre-execution financial attack warning system

Unlike traditional phishing detectors, Prometheus AI focuses on reconstructing the complete financial attack lifecycle instead of only classifying malicious content.

6. Technical Feasibility

The proposed MVP is achievable within a short development cycle using existing technologies and cloud-based AI services.

MVP Features

- Suspicious email/link submission
- Sandbox-based execution
- AiTM and BEC pattern detection
- Attack propagation simulation
- Risk dashboard and alert generation

Technology Stack

- Frontend: Next.js + TailwindCSS
- Backend: FastAPI
- Sandbox: Playwright + Docker
- Database: PostgreSQL
- AI APIs: OpenAI / Gemini

No real banking integration is required for the MVP stage.

7. Conclusion

Prometheus AI is an agentic AI cybersecurity platform designed to simulate and detect advanced financial phishing attacks before financial compromise occurs.

By combining sandbox execution, AI agents, attack propagation modeling, and explainable risk analysis, the platform provides an innovative defensive layer against modern AiTM and BEC attacks.

The system is technically feasible, ethically responsible, and highly relevant to current financial cybersecurity challenges.

8. References

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